

Quality Control Standards — Tide Motors Assembly Plant 7

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Document ID	TM7-QC-001	Revision	2
Plant	Tide Motors Assembly Plant 7	Effective Date	20 Mar 2026
Document Owner	H. Tanaka, Quality Director	Classification	Controlled — Quality

1. Purpose & Scope

This standard defines product quality control for body, paint, general assembly, and battery pack production at Tide Motors Assembly Plant 7. Aligns with the IATF 16949 QMS and the Tide Motors Global Quality Standard TGQ-12.

2. Product Specifications

Product	Key Attributes	Specification Reference
Body-in-White	Dimensional, weld quality	GD&T tolerance ± 0.5 mm key features; weld pull ≥ 4.2 kN.
Paint	Film thickness, gloss, orange peel	Topcoat 35-45 μm ; gloss 80-90 GU at 20°.
General Assembly	Torque, leak, function	Critical torques Cmk ≥ 1.67 ; door close effort ≤ 25 N.
Battery Pack	Capacity, IR, dielectric	Capacity within $\pm 2\%$ of nameplate; dielectric ≥ 500 M Ω at 500 VDC.

3. Procedures

3.1 In-Line Inspection

Vision systems on BIW, paint, and trim stations log every measurement to the MES. Defects exceeding the control limit halt the line. Containment within one cycle time.

3.2 Audit

Daily Layered Process Audit (LPA) covering safety, quality, and standardized work. CSL3 audit findings reviewed at the morning tier-2 meeting.

3.3 PPAP

Production Part Approval Process completed before launch and on any change. Suppliers maintain APQP deliverables in the Tide supplier portal.

3.4 Customer Concern

8D problem solving for any warranty repeat. Containment in 24 hours, root cause in 10 days.

4. Out-of-Specification Handling

Any OOS result triggers an immediate Hold on the affected lot, a documented investigation, and where applicable, a CAPA. Disposition decisions (Release, Rework, Reject) require QA approval. The investigation is closed in the eQMS with full traceability per the site's records retention policy.

5. Roles & Responsibilities

Quality Assurance (H. Tanaka, Quality Director) owns lot release. Operations executes per the recipe and reports deviations. Maintenance supports calibration of QC-critical instruments per the linked maintenance manual (TM7-MNT-001). The site safety procedure (TM7-SAF-001) applies to all QC laboratory activity.

6. References

- IATF 16949:2016
- AIAG-VDA FMEA Handbook
- AIAG PPAP 4th Edition
- Internal: TGQ-12, LPA-TM7-FORM-04

7. Revision History

Rev	Date	Author	Summary
1	18 Sep 2024	H. Tanaka	Initial issue.
2	20 Mar 2026	H. Tanaka	Updated specifications and SPC criteria.

Approvals

Role	Name	Signature	Date
Quality Assurance	H. Tanaka, Quality Director	_____	20 Mar 2026
Operations / Site Lead	E. Brennan, Plant Director	_____	20 Mar 2026
Regulatory / Compliance	Site Compliance Officer	_____	20 Mar 2026

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